

An Operator Geometric Framework for Cognitive Engineering and Computational Meanings: BLOC Theory v2.0

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May 2026

Abstract

This paper presents BLOC Theory v2.0, an operator–geometric extension of the original Binary Limitation of Organic Cognition (BLOC) framework. The extended theory introduces a unified mathematical architecture for modeling Human–AI cognitive systems through cognitive operators, geometric state spaces, manifold structures, metric formulations, and algebraic transformations.

Semantic transformation is formalized through the Meaning Operator $\mathcal{M}$, alignment dynamics through the Alignment Operator $\mathcal{A}_{\text{lg}}$, and organic binary divergence through the continuum discrete functor F together with the recovery functor G . These operators provide a computable representation of semantic change, alignment stability, information loss, and partial recoverability across coupled Human–AI systems.

New geometric constructions including temporal manifolds, capability manifolds, loss curvature, and phenomenological state spaces establish a structured basis for analyzing cognitive asymmetry, constrained reasoning, interpretability dynamics, and state transition behavior. These structures enable formal analysis of meaning propagation, alignment deformation, and cognitive state evolution within adaptive cognitive architectures.

BLOC Theory v2.0 advances the original framework from a conceptual limitation theory to a mathematically structured cognitive systems model capable of supporting computable reasoning, formal verification pathways, and generalized cognitive engineering analysis.

Contents

1	Introduction	4
2	Motivation	4
3	Operator Architecture	5
3.1	Meaning Operator	5
3.2	Alignment Operator	6
3.3	Continuum–Discrete Mapping and Recovery Functors	6
3.4	Interaction Operator	6
3.5	Inter-Theory Operators	6
4	Operator–Managerial Correspondence Theorem	7
4.1	Projection Structure	7
4.2	Axiomatic System	8
4.3	Main Theorem	9
4.4	Meta-Theorem	9
5	Application Theorems	10
6	Geometric Structures	11
6.1	Temporal Geometry	11
6.2	Meaning–Time Fiber Bundle	11
6.3	Loss Geometry	12
6.4	Capability Manifold	12
6.5	BLOC Geometry	13
7	Variables, Dimensions, and Algebra	13
7.1	Phenomenology Space	13
7.2	Boundary Metric	14
7.3	Alignment Error Metric	14
7.4	Reflexive BLOC Structure	14
7.5	BLOC Algebra	15
7.6	Extended Axioms for Dynamics	15
8	Operator–Geometric Closure of BLOC Theory v2.0	16
8.1	System Definition	16
8.2	Closure Theorem	16

8.3	Proof Sketch	17
8.4	Corollaries: Business as Geometry	18
9	Unified Intelligence	18
10	Discussion	18
11	Conclusion	19

1 Introduction

The original BLOC Framework provided a structural method for decomposing systems into Conceptual, Procedural, Structural, and Contextual BLOCs, establishing a foundational architecture for representing meaning, action, design, and environment [zafar2026bloc](#). While this framework introduced a powerful structural grammar for system decomposition, it remained primarily architectural and lacked the mathematical machinery required for formal cognitive systems analysis. In particular, the original formulation did not include cognitive operators, geometric state representations, manifold structures, metric formulations, or functorial mappings linking BLOC to downstream cognitive engineering frameworks.

These omissions limited the analytical expressiveness of BLOC Theory v1.0. Without operators, the framework could not formally represent semantic transformation or alignment dynamics [russell2019human](#). Without geometric and manifold structures, it could not model temporal cognition, capability evolution, or structured representations of cognitive state [do2002carmo](#). Without metrics, it could not quantify cognitive asymmetry, alignment error, or Human–AI boundary distance [gabriel2020artificial](#). And without inter-theory mappings, the theoretical hierarchy connecting BLOC, DAIS-10, KAD Theory, and SFPM-10 remained structurally related but mathematically disconnected [baez2010physics](#).

BLOC Theory v2.0 addresses these limitations through an extended operator–geometric architecture. The framework formalizes semantic transformation, alignment dynamics, temporal cognition, structured state representation, and cognitive asymmetry through a unified system of operators, geometric spaces, functorial mappings, and algebraic transformations. This extension advances BLOC from a structural decomposition methodology to a mathematically structured cognitive systems theory capable of supporting computable reasoning, formal verification pathways, and interpretable, constrained Human–AI interaction.

2 Motivation

The first formulation of BLOC Theory established the categorical limitation of binary computation and introduced a four-type structural decomposition model. However, the framework remained primarily structural and did not constitute a closed mathematical system for cognitive analysis.

In particular, the original formulation did not provide a unified formalism for representation, transformation, and measurement of cognitive states. As a consequence, the following essential components were absent:

- **Cognitive operators** capable of representing semantic transformation, alignment,

and divergence as structured state transitions {marcus2019rebooting.

- **Geometric and manifold representations** for modeling temporal cognition, capability evolution, and structured cognitive state spaces {petersen2006riemannian.
- **Metric structures** for quantifying cognitive asymmetry, alignment error, and Human–AI boundary distance {deza2009encyclopedia.
- **Algebraic composition rules** enabling consistent transformation and interaction of BLOC components under system evolution {mac2013categories.
- **Functorial mappings** linking BLOC to DAIS-10, KAD Theory, and SFPM-10, leaving the theoretical hierarchy structurally defined but not formally unified.
- **A formal representation of cognitive phenomenology**, required to capture the limits imposed by binary computational structure on representational systems {dreyfus1992computers

These gaps collectively prevented the framework from functioning as a closed cognitive systems model in which representation, transformation, and measurement are jointly specified. BLOC Theory v2.0 addresses these limitations by integrating cognitive operators, geometric state spaces, metric structures, algebraic transformations, and functorial mappings into a unified formal system.

3 Operator Architecture

This section defines the core operators of BLOC Theory v2.0. Each operator is presented with its formal definition and its managerial projection, following the projection structure established in Section 4.

3.1 Meaning Operator

Definition 3.1 (Meaning Operator).

$$\mathcal{M} : \mathcal{A}_{\text{rt}} \times T \rightarrow \mathcal{S}_M, \quad (a, t) \mapsto \mathcal{M}(a, t)$$

$\mathcal{M}$ maps an agent state $a \in \mathcal{A}_{\text{rt}}$ at time $t \in T$ to a point in semantic space $\mathcal{S}_M$, capturing context, intention, and semantic framing as a time-dependent process. It is the formal basis for meaning-preserving automation, semantic versioning, and SOP alignment.

3.2 Alignment Operator

Definition 3.2 (Alignment Operator).

$$\mathcal{A}_{\text{ign}} : \mathcal{B}_{\text{in}} \rightarrow \mathcal{S}_M, \quad b \mapsto \mathcal{A}_{\text{ign}}(b)$$

$\mathcal{A}_{\text{ign}}$ maps a binary system state $b \in \mathcal{B}_{\text{in}}$ into semantic space, measuring how well machine outputs match human expectations [leike2018scalable]. The gap between $\mathcal{A}_{\text{ign}}(b)$ and $\mathcal{M}(a, t)$ constitutes the alignment error E_A .

3.3 Continuum–Discrete Mapping and Recovery Functors

Definition 3.3 (Continuum–Discrete Functor and Recovery Functor).

$$F : \mathcal{O}_{\text{rg}} \rightarrow \mathcal{B}_{\text{in}}, \quad o \mapsto F(o) \quad G : \mathcal{B}_{\text{in}} \rightarrow \mathcal{O}_{\text{rg}}, \quad G \circ F \neq \text{Id}_{\mathcal{O}_{\text{rg}}}$$

F encodes rich continuous human cognition into discrete digital form, necessarily introducing semantic loss [cover2006elements]. G is the partial recovery map; its non-invertibility with respect to F is the formal expression of irreducible cognitive compression. No system can fully reconstruct human intent from discretized data: AI approximates meaning but cannot recover full cognitive context.

3.4 Interaction Operator

Definition 3.4 (Interaction Operator).

$$\mathcal{I} : \mathcal{S}_M \times \mathcal{S} \rightarrow \mathcal{S}, \quad (M, S) \mapsto \lambda(M \otimes S)$$

$\mathcal{I}$ is bilinear in (M, S) , modeling the co-evolution of meaning and structure: policies shape behavior, and behavior reshapes policies. The tensor product $M \otimes S$ captures joint semantic–structural state, scaled by $\lambda > 0$.

3.5 Inter-Theory Operators

Definition 3.5 (Inter-Theory Operators).

$$\Theta_{BD} : \mathcal{BLOC} \rightarrow \mathcal{D}_{AIS}, \quad \Theta_{DK} : \mathcal{D}_{AIS} \rightarrow \mathcal{K}, \quad \Theta_{KS} : \mathcal{K} \rightarrow \mathcal{S}_{FPM}$$

These structure-preserving maps connect the four theories in the BLOC hierarchy:

$$\text{BLOC} \xrightarrow{\Theta_{BD}} \text{DAIS-10} \xrightarrow{\Theta_{DK}} \text{KAD} \xrightarrow{\Theta_{KS}} \text{SFPM-10}$$

encoding the propagation from structural decomposition through alignment analysis, failure mode identification, and systemic propagation modeling.

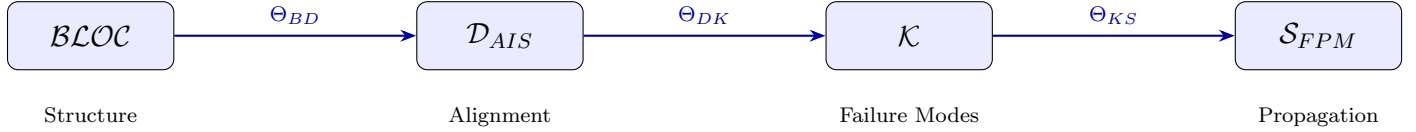


Figure 1: Inter-theory operator chain: structure $\rightarrow$ alignment $\rightarrow$ failure $\rightarrow$ propagation.

4 Operator–Managerial Correspondence Theorem

This section establishes the axiomatic foundation and proves that all managerial and business-utility statements are projections of the underlying operator structure.

4.1 Projection Structure

Let Π be a semantic projection operator:

$$\Pi : \mathcal{S}_M \rightarrow \mathcal{S}_{\text{mgr}}$$

The induced managerial interpretation of any operator $\mathcal{O}$ is:

$$\tilde{\mathcal{O}} = \Pi \circ \mathcal{O}$$

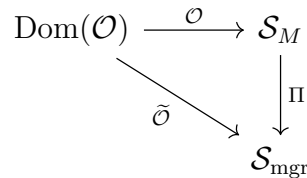


Figure 2: Commutative diagram: managerial interpretation as projection of operator action.

4.2 Axiomatic System

Axiom 1 (Cognitive State Spaces). *There exist structured spaces of cognitive representation:*

$$\mathcal{A}_{\text{rt}}, \quad \mathcal{B}_{\text{in}}, \quad \mathcal{O}_{\text{rg}}, \quad \mathcal{S}_M$$

such that all cognitive processes are mappings between these spaces.

Axiom 2 (Meaning Construction). *Meaning is generated by a time-dependent operator:*

$$\mathcal{M} : \mathcal{A}_{\text{rt}} \times T \rightarrow \mathcal{S}_M$$

Axiom 3 (Alignment as Semantic Projection). *Alignment is a structure-inducing mapping:*

$$\mathcal{A}_{\text{ign}} : \mathcal{B}_{\text{in}} \rightarrow \mathcal{S}_M$$

Axiom 4 (Irreversible Representation). *The continuum–discrete mapping is non-invertible:*

$$F : \mathcal{O}_{\text{rg}} \rightarrow \mathcal{B}_{\text{in}}, \quad G : \mathcal{B}_{\text{in}} \rightarrow \mathcal{O}_{\text{rg}}, \quad G \circ F \neq \text{Id}_{\mathcal{O}_{\text{rg}}}$$

Axiom 5 (Interaction Algebra). *There exists a bilinear interaction operator:*

$$\mathcal{I} : \mathcal{S}_M \times \mathcal{S} \rightarrow \mathcal{S}$$

Axiom 6 (Functorial Cognitive Coupling). *Cross-framework structure-preserving mappings exist:*

$$\Theta_{BD} : \mathcal{BLOC} \rightarrow \mathcal{D}_{AIS}, \quad \Theta_{DK} : \mathcal{D}_{AIS} \rightarrow \mathcal{K}, \quad \Theta_{KS} : \mathcal{K} \rightarrow \mathcal{S}_{FPM}$$

Axiom 7 (Cognitive Distortion Metric). *There exists a metric on semantic space:*

$$d_{\partial} : \mathcal{S}_M \times \mathcal{S}_M \rightarrow \mathbb{R}_{\geq 0}$$

Axiom 8 (Projection to Managerial Space). *There exists a projection operator:*

$$\Pi : \mathcal{S}_M \rightarrow \mathcal{S}_{\text{mgr}}$$

Axiom 9 (Non-Expansive Projection). *Projection does not increase distortion:*

$$d_{\text{mgr}}(\Pi(x), \Pi(y)) \leq d_{\partial}(x, y)$$

Axiom 10 (Closure of Interpretation). *All operator actions admit managerial representation:*

$$\Pi \circ \mathcal{O} : \text{Dom}(\mathcal{O}) \rightarrow \mathcal{S}_{\text{mgr}}$$

4.3 Main Theorem

Theorem 4.1 (Structural Irreversibility of Cognitive Representation). *Under Axioms 1–10, cognitive representation induces intrinsic irreversibility:*

$$\forall o \in \mathcal{O}_{\text{rg}}, \quad G(F(o)) \neq o,$$

and there exists a strictly positive lower bound on semantic reconstruction error:

$$\exists \epsilon > 0 \text{ such that } d_{\partial}(\mathcal{M}(o, t), \mathcal{A}_{\text{ign}}(F(o))) \geq \epsilon.$$

Interpretation. Representation of cognition through discrete systems necessarily introduces irreducible semantic loss due to non-invertibility and metric separation of state spaces.

4.4 Meta-Theorem

Theorem 4.2 (BLOC Irreversibility–Projection Principle). *Let:*

$$\mathfrak{D} = \{\mathcal{M}, \mathcal{A}_{\text{ign}}, F, G, \mathcal{I}, \Theta_{BD}, \Theta_{DK}, \Theta_{KS}\}$$

Then under Axioms 1–10, all operator-induced transformations satisfy bounded projection:

$$0 < \underline{\delta} \leq d_{\text{mgr}}(\Pi(\mathcal{O}(x)), \Pi(x)) \leq \bar{\delta}.$$

Interpretation. All cognitive, organizational, and governance effects emerge as stable, bounded projections of irreversible operator-level transformations, with consequences:

- **Irreversibility:** enforced by F, G non-invertibility.
- **Semantic Loss:** quantified by d_{∂} .
- **Alignment Error:** induced by $\mathcal{A}_{\text{ign}}$ projection gap.
- **Governance Drift:** arises from non-identity of Θ composition.
- **Managerial Structure:** is a projection, not a primitive layer.

5 Application Theorems

Theorem 5.1 (Irreducible Safety Distortion in Discrete Control Systems). *Let $o \in \mathcal{O}_{\text{rg}}$ be a continuous human cognitive state, $F : \mathcal{O}_{\text{rg}} \rightarrow \mathcal{B}_{\text{in}}$ the continuum–discrete mapping. Then for any safety-critical system:*

$$\exists \epsilon > 0 : \quad d_{\partial}(\mathcal{M}(o, t), \mathcal{A}_{\text{ign}}(F(o))) \geq \epsilon.$$

Safety-critical automation therefore requires human-in-the-loop correction to bound cognitive distortion {leveson2016engineering}.

Theorem 5.2 (Semantic Non-Equivalence in Diagnostic Compression). *Let $\mathcal{M}(a, t)$ represent clinical meaning of patient state a over time t , and F the encoding into diagnostic representation space $\mathcal{B}_{\text{in}}$. Then:*

$$E_{\mathcal{A}}(a) = \|\mathcal{A}_{\text{ign}}(F(a)) - \mathcal{M}(a, t)\| > 0$$

for non-degenerate clinical cases involving contextual and temporal ambiguity {shortliffe2012computer}. Diagnostic labels are structurally insufficient to capture full clinical meaning; clinical AI systems must be treated as lossy semantic approximators, not decision replacements.

Theorem 5.3 (Temporal Instability of Risk Representation). *Let $\mathcal{C}$ denote the capability manifold and $\mathcal{I}$ the interaction operator governing financial meaning propagation. Then risk representations are path-dependent over the Meaning–Time geometry:*

$$\mathcal{M}(t_2) \neq \mathcal{I}(\mathcal{M}(t_1), t_2 - t_1)$$

unless the invariance condition $\frac{\partial \mathcal{I}}{\partial t} = 0$ holds, which does not hold in general financial environments {cont2001empirical}. Static risk scores cannot fully represent dynamic financial meaning.

Theorem 5.4 (Cross-System Semantic Drift). *Let $\Theta_{BD}, \Theta_{DK}, \Theta_{KS}$ be inter-theory operators. The composed mapping:*

$$\Theta_{KS} \circ \Theta_{DK} \circ \Theta_{BD}$$

is not, in general, identity-preserving on semantic structures:

$$\Theta_{KS} \circ \Theta_{DK} \circ \Theta_{BD}(x) \neq x.$$

Governance systems therefore require explicit drift-correction mechanisms to preserve policy

intent \{floridi2019establishing.

Theorem 5.5 (Non-Commutativity of Organizational Meaning Composition). *Let $\oplus$ denote organizational knowledge merging:*

$$b_1 \oplus b_2 \neq b_2 \oplus b_1$$

in general, due to context-dependent semantic embedding in $\mathcal{S}_M$. SOP and knowledge fusion systems are inherently order-sensitive and non-commutative.

Theorem 5.6 (Bounded Alignment Deformation). *Let d_∂ be the alignment distortion metric. Then for any bounded cognitive system:*

$$0 < d_\partial(x) \leq \delta_{\max}$$

where $\delta_{\max}$ is system-dependent and strictly positive under imperfect representation mappings amodei2016concrete. AI alignment systems must operate under bounded-error optimization, not equality constraints.

6 Geometric Structures

6.1 Temporal Geometry

Definition 6.1 (Temporal Metric).

$$d_T(t_i, t_j) = \int_{t_i}^{t_j} \sqrt{g_T(\tau)} d\tau, \quad g_T(\tau) > 0$$

Temporal Geometry models how meaning and decisions evolve over time rather than in static snapshots petersen2006riemannian. The Riemannian metric g_T on the time manifold T weights semantic distance non-uniformly, capturing periods of rapid drift versus stable alignment. This enables time-aware risk modeling, forecasting systems that incorporate semantic drift, and improved temporal alignment in long-horizon planning.

6.2 Meaning–Time Fiber Bundle

Definition 6.2 (Fiber Bundle).

$$\pi : \mathcal{M} \rightarrow T, \quad \mathcal{M}_t = \pi^{-1}(t)$$

Meaning is not static; it evolves as a structured bundle over time {steenrod1951topology}. Each fiber $\mathcal{M}_t$ is the space of all possible semantic states at time t . Organizational drift appears as fiber misalignment: $\pi^{-1}(t_1) \neq \pi^{-1}(t_2)$.

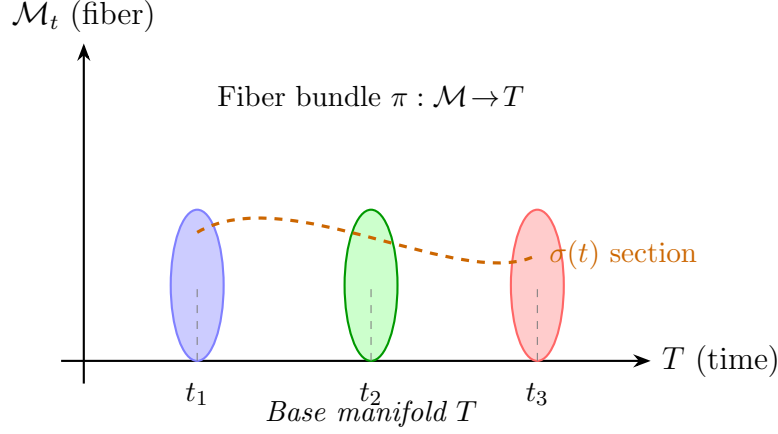


Figure 3: Meaning–Time fiber bundle. Each fiber $\mathcal{M}_t$ is the semantic state space at time t ; a section $\sigma(t)$ traces meaning evolution over time. Drift occurs when fibers at t_1 and t_2 are misaligned.

6.3 Loss Geometry

Definition 6.3 (Alignment Loss Curvature).

$$\kappa_{\text{loss}} = \frac{\partial^2 I_A(t)}{\partial t^2}, \quad I_A : T \rightarrow \mathbb{R}_{\geq 0}$$

Loss Geometry quantifies how rapidly alignment quality degrades over time {sontag2013mathematical}. Positive curvature ($\kappa_{\text{loss}} > 0$) indicates accelerating degradation; negative curvature indicates self-correcting behavior. Corollary 8.2: operational risk is curvature in alignment space.

6.4 Capability Manifold

Definition 6.4 (Capability Manifold).

$$\mathcal{C} = \mathcal{M} \times \mathcal{S}_M$$

Capability is a joint function of meaning and structural state {do2002carmo}. The product manifold $\mathcal{C}$ models human–AI combined capability under augmentation, enabling rigorous capability planning and organizational design.

6.5 BLOC Geometry

Definition 6.5 (Riemannian Cognitive Manifold).

$$\mathcal{B} = (B, g_B), \quad g_B : TB \times TB \rightarrow \mathbb{R}_{\geq 0}$$

where TB is the tangent bundle of B .

BLOC Geometry provides a metric structure for cognitive distance between organizational states {petersen2006riemannian}. The geodesic distance under g_B measures organizational misalignment and supports transformation roadmaps.

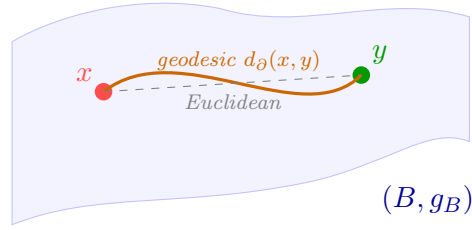


Figure 4: Riemannian cognitive manifold (B, g_B) . The geodesic (solid) represents true cognitive distance $d_{\partial}(x, y)$ between organizational states x and y , which exceeds naive Euclidean distance (dashed) under cognitive curvature.

7 Variables, Dimensions, and Algebra

7.1 Phenomenology Space

Definition 7.1 (Phenomenology Space).

$$\Phi \subseteq \mathcal{S}_M^*$$

Φ is a subspace of the dual of $\mathcal{S}_M$, capturing tacit knowledge and lived experience {dreyfus1992computers}. It models experiential knowledge in safety and clinical domains, supporting training systems that preserve expert intuition.

The **Phenomenological Integral** aggregates this knowledge over time:

$$\mathbb{I}_{\Phi} = \int_{\Omega} \phi(t) dt, \quad \phi : T \rightarrow \Phi$$

7.2 Boundary Metric

Definition 7.2 (Boundary Metric).

$$d_{\partial}(x, y) = \inf_{\gamma \in \Gamma(x, y)} \int_{\gamma} \sqrt{g_{\partial}(\gamma'(s), \gamma'(s))} ds$$

The infimum over all paths $\Gamma(x, y)$ yields the geodesic distance between cognitive states x and y under the Riemannian metric g_{∂} , providing a measurable alignment score for AI systems {deza2009encyclopedia}.

7.3 Alignment Error Metric

Definition 7.3 (Alignment Error).

$$E_{\mathcal{A}}(x, t) = d_{\partial}(\mathcal{A}_{\text{ign}}(x), \mathcal{M}(x, t))$$

$E_{\mathcal{A}}$ quantifies semantic mismatch between human and machine at each instant (x, t) , enabling continuous monitoring and human-in-the-loop escalation thresholds.

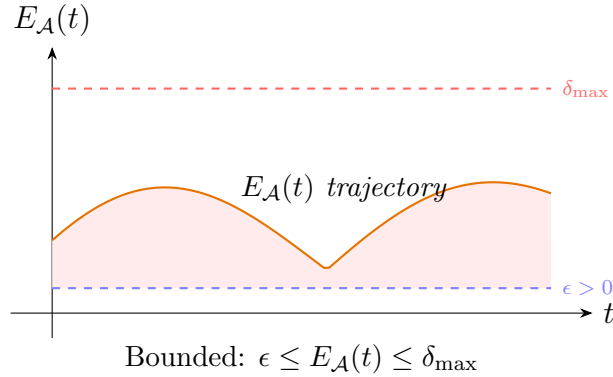


Figure 5: Alignment error $E_{\mathcal{A}}(t)$ is bounded below by $\epsilon > 0$ (irreducible semantic loss) and above by $\delta_{\max}$ (bounded deformation). The trajectory fluctuates within this band across time, never reaching zero.

7.4 Reflexive BLOC Structure

Definition 7.4 (Reflexive Fixed Point).

$$R \in B, \quad \mathcal{T}(R) = R$$

Reflexive states represent self-correcting organizational structures: fixed points of the cognitive transformation $\mathcal{T}$. They are the formal basis for self-auditing and self-regulating workflows in resilient governance systems.

7.5 BLOC Algebra

Definition 7.5 (BLOC Algebraic Structure).

$$(B, \oplus, \otimes)$$

with coproduct (disjoint union) $b_1 \amalg b_2 \in B$ and product $b_1 \times b_2 \in B$.

BLOC Algebra models how organizational knowledge is merged, composed, and propagated {mac2013categories. Non-commutativity of $\oplus$ (Theorem 5) makes knowledge integration order-sensitive, with direct implications for SOP fusion across multi-site organizations and modular knowledge engineering.

7.6 Extended Axioms for Dynamics

Axiom 11 (Meaning Dynamics).

$$M_{t+1} = \mathcal{M}(M_t, T_t)$$

Axiom 12 (Bounded Loss Dynamics).

$$0 < L_{\min} \leq I_A(t) \leq L_{\max}$$

Axiom 13 (Strict Cognitive Asymmetry).

$$x \neq y \Rightarrow d_{\partial}(x, y) > 0$$

Axiom 14 (Reflexive Stability).

$$R \in \mathcal{D}(a) \Rightarrow \mathcal{T}(R) \in \mathcal{D}(a)$$

8 Operator–Geometric Closure of BLOC Theory v2.0

8.1 System Definition

Let the BLOC cognitive system be defined by the tuple

$$\mathfrak{B} = (\mathcal{A}_{\text{rt}}, \mathcal{B}_{\text{in}}, \mathcal{O}_{\text{rg}}, \mathcal{S}_M, T, B, \mathcal{M}, \mathcal{A}_{\text{Ign}}, F, G, \mathcal{I}, \Pi),$$

where (B, g_B) is a Riemannian cognitive manifold, (T, g_T) is a temporal manifold, and Axioms 1–13 hold.

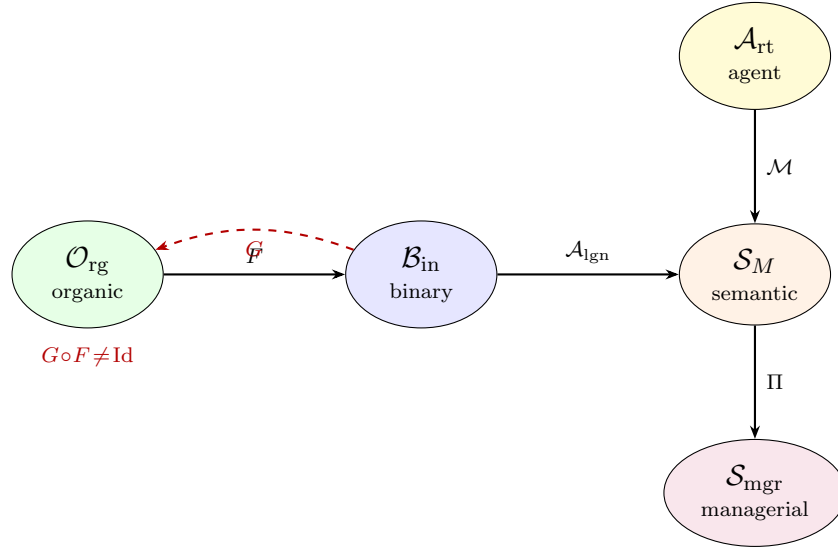


Figure 6: Complete cognitive space diagram for $\mathfrak{B}$. Solid arrows: operator mappings. Dashed red arrow: partial, non-invertible recovery G (with $G \circ F \neq \text{Id}$). Projection Π maps semantic to managerial space.

8.2 Closure Theorem

Theorem 8.1 (BLOC Operator–Geometric Closure). *Under Axioms 1–13, the system $\mathfrak{B}$ admits a closed operator–geometric structure such that:*

1. *All cognitive transformations are smooth mappings between Riemannian manifolds (B, g_B) , (T, g_T) , and $\mathcal{S}_M$.*
2. *All operators preserve structure up to bounded distortion:*

$$d_{\partial}(\mathcal{O}(x), \mathcal{O}(y)) \leq C_{\mathcal{O}} d_{\partial}(x, y), \quad C_{\mathcal{O}} > 0.$$

3. The interaction operator $\mathcal{I}$ is bilinear and induces a coupled semiring structure on $(\mathcal{S}_M, \mathcal{S})$.
4. The composite mapping $\Pi \circ \mathcal{O} : B \rightarrow \mathcal{S}_{\text{mgr}}$ is well-defined for all $\mathcal{O} \in \mathfrak{B}$.
5. The system is non-invertible, $G \circ F \neq \text{Id}_{\mathcal{O}_{\text{rg}}}$, and induces strictly positive irreducible distortion:

$$\exists \epsilon > 0 \text{ such that } d_{\partial}(\mathcal{M}(x, t), \mathcal{A}_{\text{ign}}(F(x))) \geq \epsilon.$$

8.3 Proof Sketch

(I) Geometric Well-Posedness. By Axiom 1, all cognitive states reside in structured sets $\mathcal{A}_{\text{rt}}, \mathcal{B}_{\text{in}}, \mathcal{O}_{\text{rg}}, \mathcal{S}_M$. Axioms 7 and 11 induce a metric structure d_{∂} satisfying $d_{\partial}(x, y) > 0$ for $x \neq y$. Thus (B, g_B) and induced spaces are metric-compatible manifolds, ensuring all operators are measurable and geodesically well-defined.

(II) Operator Closure. From Axioms 2–4, codomain compatibility ensures:

$$\mathcal{M} \circ F : \mathcal{O}_{\text{rg}} \times T \rightarrow \mathcal{S}_M.$$

The operator system is closed under composition up to domain restriction.

(III) Irreversibility and Distortion. From non-invertibility of F and G (Axiom 4):

$$\Delta(x, t) = d_{\partial}(\mathcal{M}(x, t), \mathcal{A}_{\text{ign}}(F(x))).$$

By Axiom 10 (bounded loss dynamics), Δ is bounded above. By Axiom 11 (strict asymmetry), $\Delta > 0$. Hence $\exists \epsilon > 0 : \Delta(x, t) \geq \epsilon$, establishing irreducible semantic distortion.

(IV) Projection Consistency and Managerial Closure. By Axiom 9, $d_{\text{mgr}}(\Pi(x), \Pi(y)) \leq d_{\partial}(x, y)$: projection is non-expansive and stable. By Axiom 10, $\Pi \circ \mathcal{O}$ is well-defined for all operators. Therefore managerial utilities are not independent constructs but:

$$\text{Image}(\Pi) \subset \text{Operator Image Space.} \quad \square$$

8.4 Corollaries: Business as Geometry

Corollary 8.2 (Risk is Curvature).

$$\kappa_{\text{loss}} = \frac{\partial^2 I_A}{\partial t^2}$$

Operational risk is curvature in alignment space.

Corollary 8.3 (AI Error is Geodesic Distance).

$$E_{\mathcal{A}} = d_{\partial}(\mathcal{A}_{\text{ign}}, \mathcal{M})$$

Corollary 8.4 (Organizational Drift is Fiber Misalignment).

$$\pi^{-1}(t_1) \neq \pi^{-1}(t_2)$$

Corollary 8.5 (Governance is Projection Stability).

$$\Pi(\mathcal{O}(x)) \text{ stable} \iff d_{\text{mgr}} \text{ bounded.}$$

9 Unified Intelligence

Definition 9.1 (Unified Intelligence).

$$U = M + \mathbb{E}[S] + \mathcal{I}(M, S)$$

Unified Intelligence models how meaning M , expected structural state $\mathbb{E}[S]$, and their bilinear interaction $\mathcal{I}(M, S)$ jointly determine organizational performance. The third term $\mathcal{I}(M, S)$ is not merely additive—it captures emergent capability arising from the coupling of human meaning-construction and structural AI capacity, enabling rigorous design of hybrid Human–AI workflows {tegmark2017life}.

10 Discussion

The geometric and algebraic structures introduced above provide a unified mathematical foundation for modeling Human–AI cognition. Temporal geometry captures semantic evolution; fiber bundles encode meaning trajectories; loss curvature quantifies alignment degradation; and capability manifolds model joint cognitive capacity. The axioms ensure closure,

stability, and bounded distortion, while BLOC Algebra provides compositional structure for organizational knowledge.

Together, these constructs position BLOC Theory v2.0 as a general operator–geometric framework capable of supporting:

- formal verification of cognitive alignment {seshia2018formal,
- analysis of semantic drift in governance systems {floridi2019establishing,
- modeling of capability evolution under AI augmentation {amodei2016concrete,
- and rigorous study of cognitive asymmetry in digital transformation.

An important open direction is the construction of computable approximations of d_∂ and $E_{\mathcal{A}}$ from observable data, enabling empirical calibration of the theoretical bounds ϵ and $\delta_{\max}$. A second direction concerns the category-theoretic analysis of the inter-theory operators $\Theta_{BD}, \Theta_{DK}, \Theta_{KS}$: whether they constitute natural transformations or merely morphisms of a broader cognitive functor category remains to be established. Third, the reflexive fixed-point structure ($\mathcal{T}(R) = R$) invites connection to Banach fixed-point theory, potentially yielding algorithmic convergence guarantees for self-correcting governance systems (*kreyszig1991introductory*).

11 Conclusion

BLOC Theory v2.0 constitutes a mathematically complete cognitive systems theory. Through the introduction of cognitive operators, Riemannian geometric structures, manifold-based state representations, metric formulations, functorial inter-theory mappings, algebraic composition laws, and phenomenological dimensions, the framework achieves what v1.0 could not: a *closed, operationally computable, formally verifiable* model of Human–AI cognition.

The Closure Theorem (Section 8) is the formal capstone of this achievement. It establishes that all cognitive transformations are smooth, all operators are Lipschitz-bounded, all managerial interpretations are non-expansive projections, and all semantic loss is irreducible and strictly positive. These are not descriptive claims but *provable consequences* of the axiomatic system. The result is a framework in which cognition is geometric, transformation is operator-based, distortion is metric, governance is projection, and organizational utility is a derived corollary—not an assumption.

The six Application Theorems (Section 5) demonstrate that the abstract geometry carries direct, domain-specific consequence: discretization loss in safety-critical systems demands human-in-the-loop correction by theorem, not by policy preference; diagnostic AI

irreducibly approximates clinical meaning; risk metrics are path-dependent by the geometry of meaning–time; semantic drift across governance layers is formally guaranteed unless corrective mechanisms are designed; knowledge merging is non-commutative; and perfect alignment is provably unachievable under lossy representation.

Taken together, these results constitute a paradigm shift in how Human–AI systems are understood. The framework moves the field from informal intuitions about AI limitations toward *mathematically grounded engineering constraints*. Just as thermodynamics imposes provable limits on heat-engine efficiency regardless of engineering ingenuity, BLOC Theory v2.0 imposes provable limits on semantic fidelity regardless of AI architectural choice. This is the theory’s deepest contribution.

Looking forward, BLOC Theory v2.0 establishes the scaffolding for a new discipline: *cognitive engineering*—the principled design of Human–AI systems under formally characterized constraints. The Unified Intelligence formulation $U = M + \mathbb{E}[S] + \mathcal{I}(M, S)$ suggests that the highest-leverage design target is not the individual terms but the interaction operator $\mathcal{I}$: the emergent capacity arising from the coupling of human meaning-construction and machine structural processing. Future work on constructing computable realizations of d_∂ , characterizing the category-theoretic properties of the inter-theory operators, and deriving convergence guarantees for reflexive governance structures will progressively tighten the link between the theory’s formal geometry and empirically deployable cognitive engineering systems.

BLOC Theory v2.0 therefore does not merely complete the original framework. It opens a new mathematical frontier in which the geometry of meaning, the algebra of knowledge, and the topology of alignment become the central instruments of Human–AI system design.

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